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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

HACKATHON 2026

Idea Submission Template — Code. Create. Innovate.

Team Name:

Team Raksha

Problem Statement No:

2

Problem Statement Title:

Improving Safety at Large Public Events

Team Leader Name:

[Your Name]

IDEA TITLE

Know the crowd. Know the person. Stop the surge.

Live tracking plus instant person files.

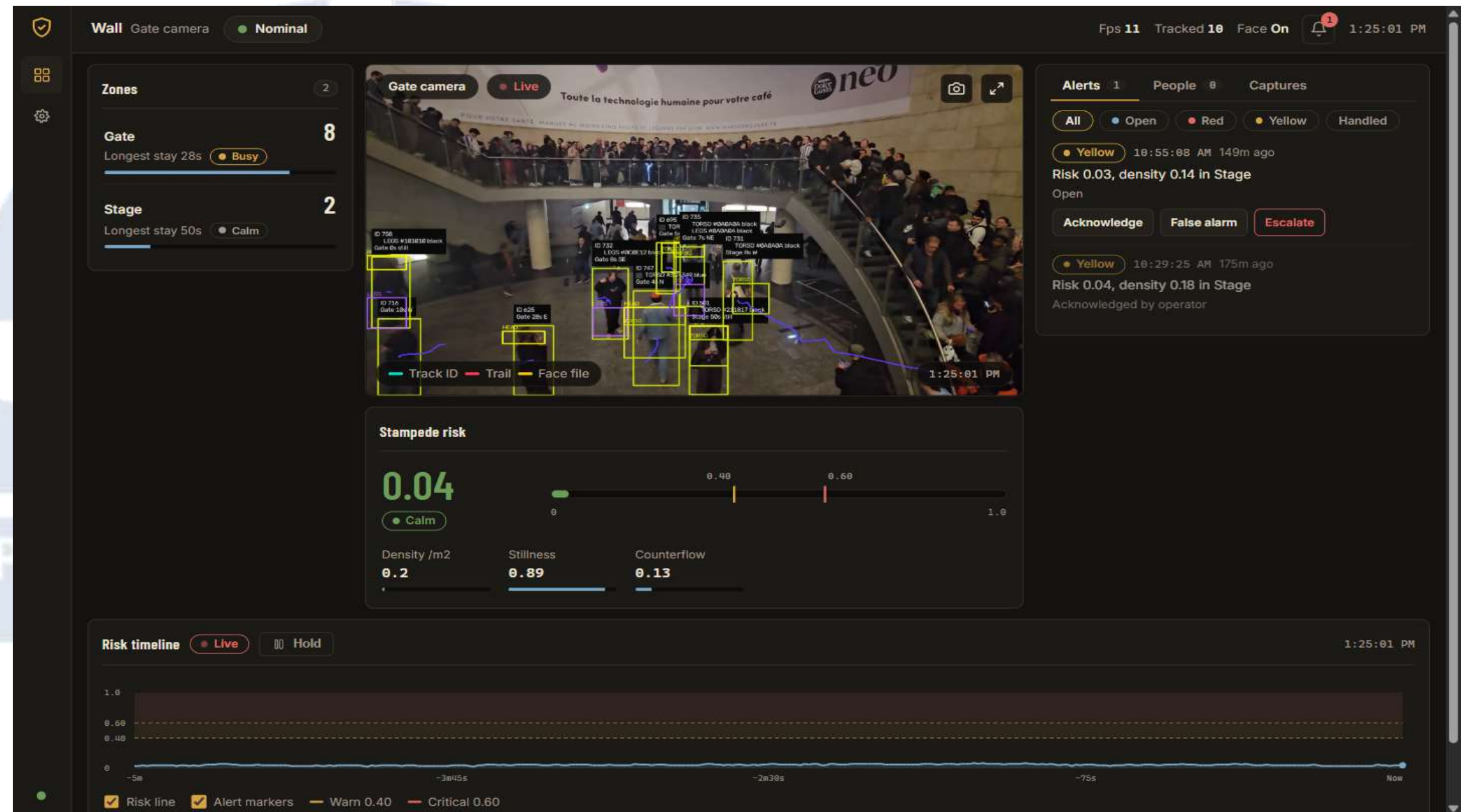
Surge warnings before the crush.

0.4s <1s 15

face to file pop yellow to guard fps live wall

Watch it work:

1. A face appears on camera.
2. Their file pops up in 0.4s.
3. A crowded zone turns yellow in under 1s.

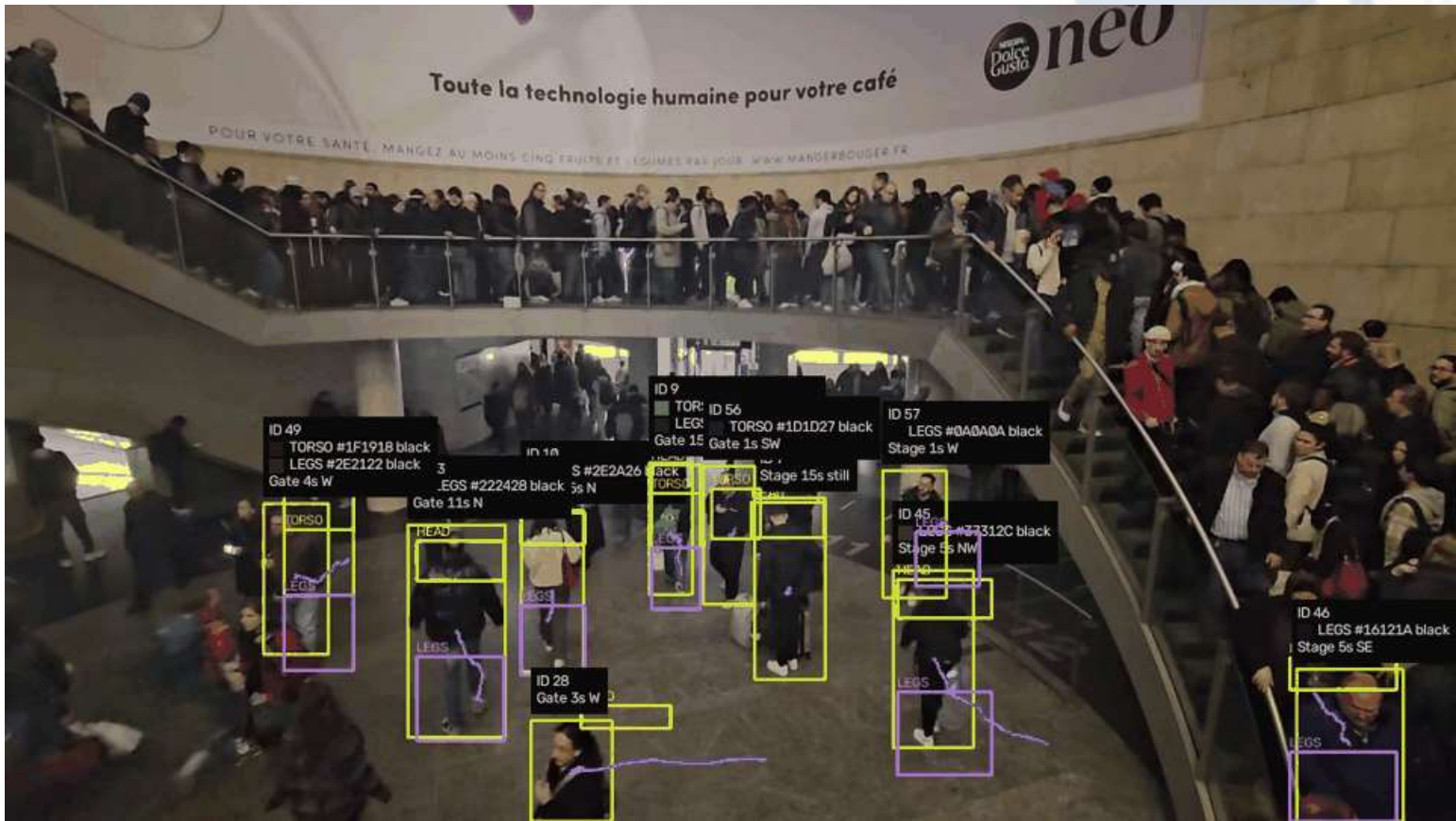


Above: the live wall on real station footage.

TECHNICAL APPROACH



In plain words: the camera watches, the software follows every person, and it shouts before a crush builds.



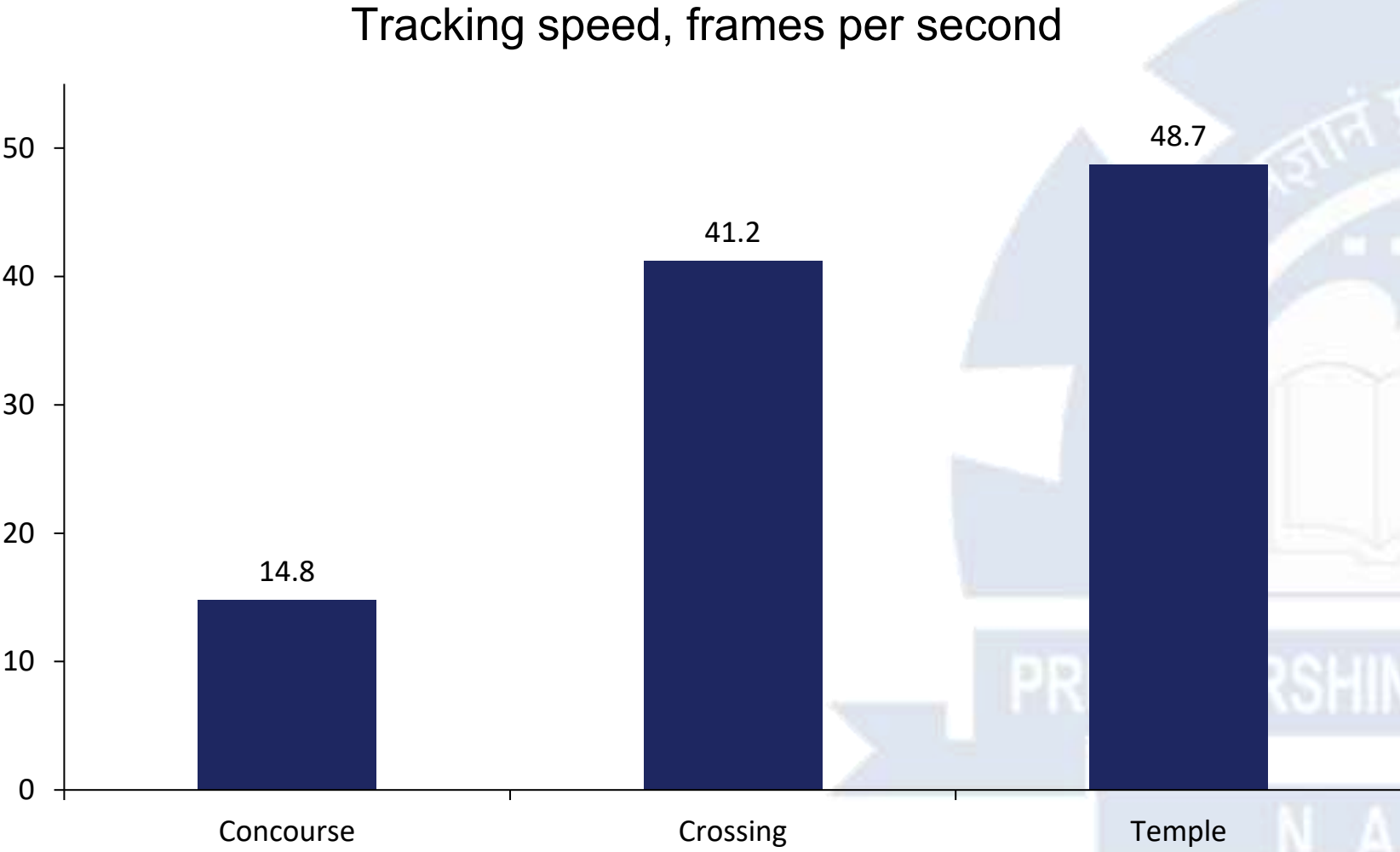
Quality gate

Only sharp, front-facing photos join the gallery. Blurry faces are ignored, so matches stay honest.

Measured

Peak 18 people tracked at once. 850 IDs held across 2 minutes of station video.

FEASIBILITY AND VIABILITY



One RTX 4050 laptop. Concourse is the heaviest crowd and it still runs smooth.

61 ms

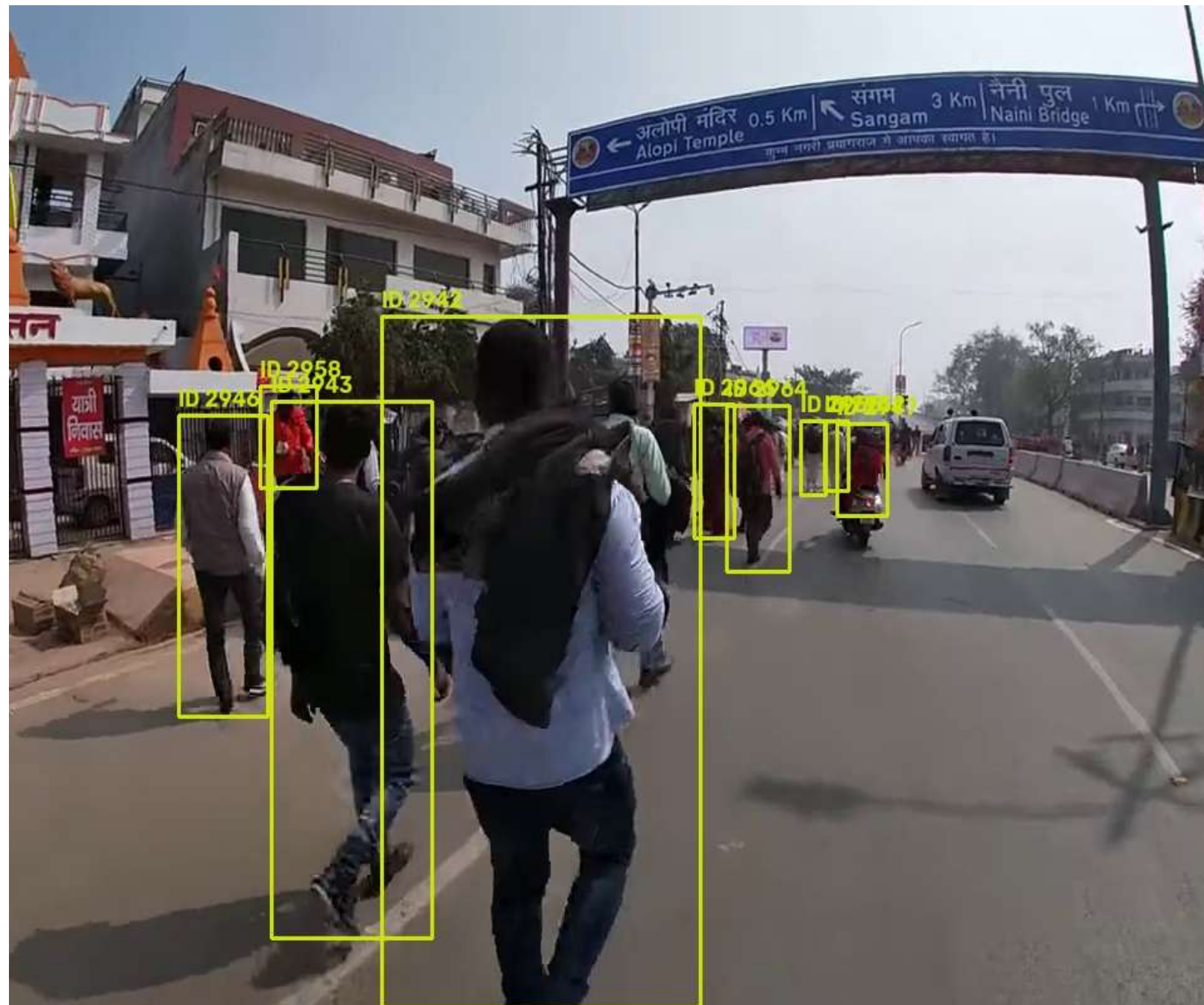
face match on GPU, was 438 ms on CPU

1.2+

risk score where RED fired on dense test



IMPACT AND BENEFITS



Before: 50 screens, one tired operator, nothing flagged.

After: one wall watches everything and pushes only the risk to guards.

Organizers

Bottleneck flagged 60s before humans see it. Guards get a zone-wise push.

Police

Find a lost child or suspect by photo in under 2s. Dossier plus trail in one click.

Public

Safer exits, faster help. Faces stay blurred in storage.

RESEARCH AND REFERENCES

YOLOv8 plus ByteTrack, 2025-26 studies: best speed balance; ByteTrack 14 percent faster than BoT-SORT.

ArcFace buffalo plus Faiss: 99.5 percent LFW, cosine 0.40, vote smoothing steadies matches.

Stampede models: AURORA hybrid, FTLE bottlenecks, Delhi pilot cut response 37 percent.

India context: Hathras, Kumbh, station crushes come from density plus blocked exits.

Blurred storage

faces blurred on disk

7-day purge

auto delete, no backlog

Audit log

hash-chained actions

No ID linkage

opt-in gallery only